



The Circle of Para-Xylene

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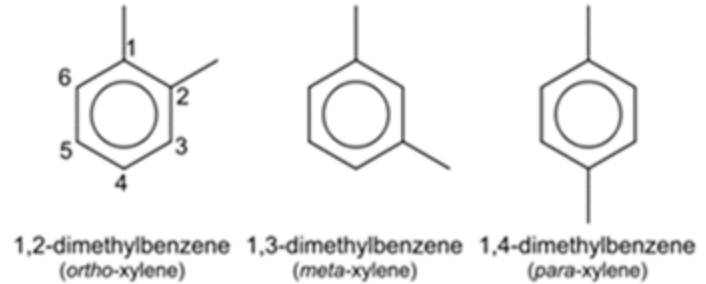




Introduction

What are Xylenes?

- Derived from crude oil
- Colorless
- Flammable
- Sweet Aroma
- Aromatic hydrocarbon Compound



Molecular structures of xylene isomers

Isomer

- Ortho-
- Meta-
- Para-



WHY Para-xylene?

WHY para-Xylene

PX → PTA and DMT → PET plastics and polyester

The Good

- Chinese textile industry
- New Available Technology

The Bad

- Overly produced
- Production Rate under 80% in the 20th century

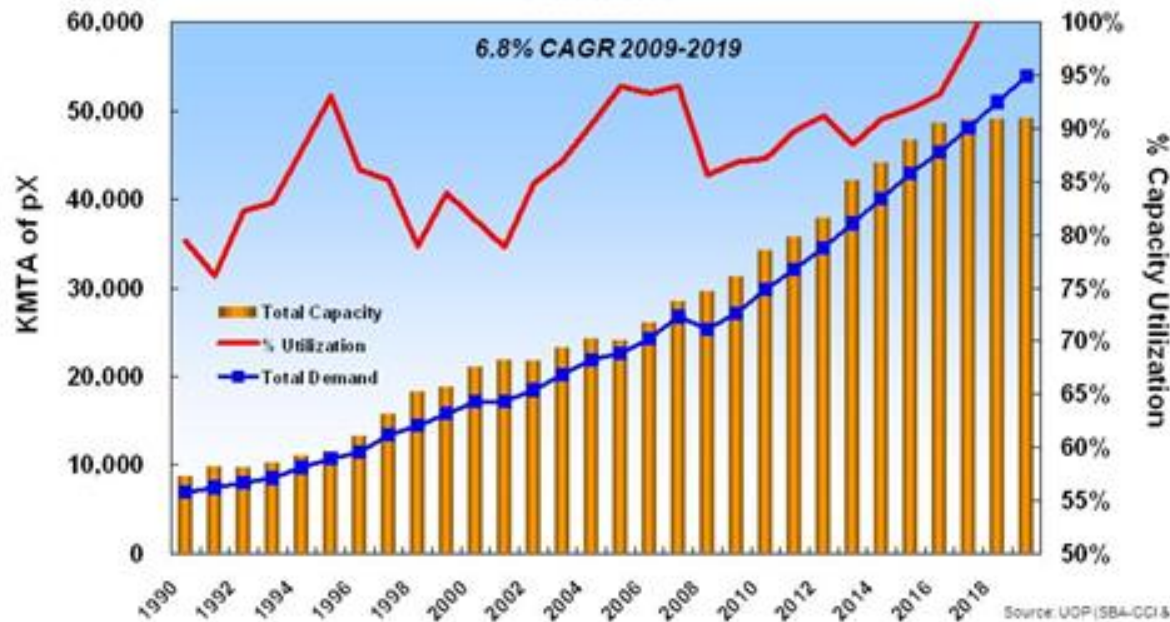
The Beautiful

- Market balanced
- Growth
- Time gap allows more stable Market

Global pX Supply / Demand

Latest View - Awarded Capacity

October 2010



Polyester (pX) Demand Growth Driven By GDP & Emerging "Middle Class"



Production of *para*-xylene

Possible Production Solutions

Toluene Methanol Alkylation

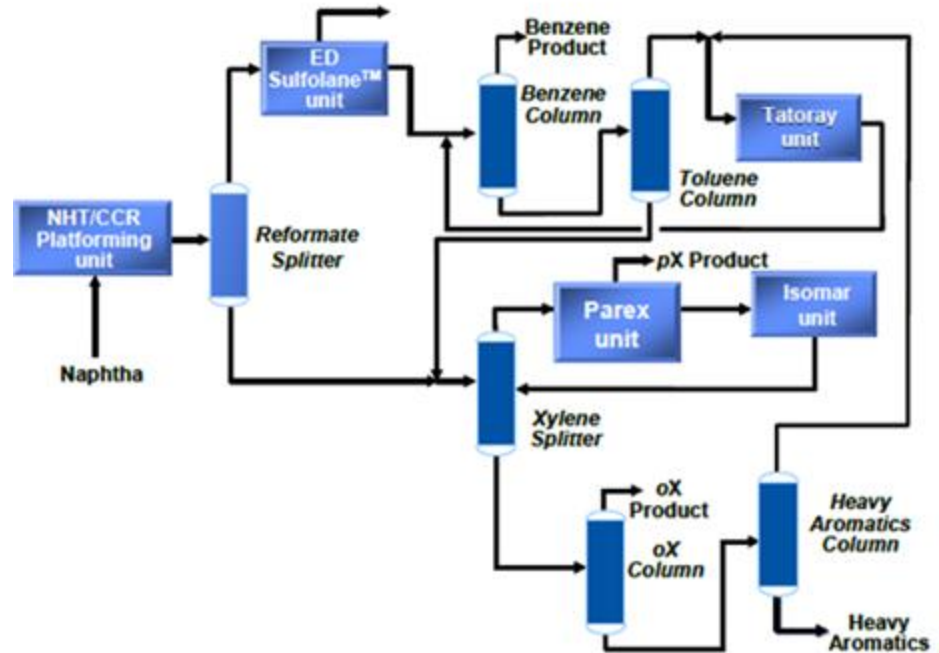
- i. Produces xylenes from toluene and methanol reacted over a catalyst to induce alkylation. [3]
- ii. Yield between 20% and 35% [3]
- iii. Pros: crude oil not required. [3]
- iv. Cons: low yield of product with respect to costly inputs. [17]

UOP PAREX Process [19]

- i. Separates a naptha feedstock into benzene, toluene, and *para*-xylene from heavier aromatics.
- ii. The product from the separator is 99.5%+ pure *para*-xylene and the separator.
- iii. Pros: produces very pure product and operates at a relatively cool temperature.
- iv. Cons: requires naptha feedstock from crude oil.

Chosen Process: UOP PAREX Separation with ADS-47 Adsorbent

- Produces high purity *para*-xylene (99.7%). [19]
- Isomerization re-equilibrates C8 aromatics to obtain maximum production from feedstock. [19]
- Creates biproducts that can be sold rather than used. [17]
- Can use existing equipment to upgrade to new enhanced process; Increases production by up to 30% [20]





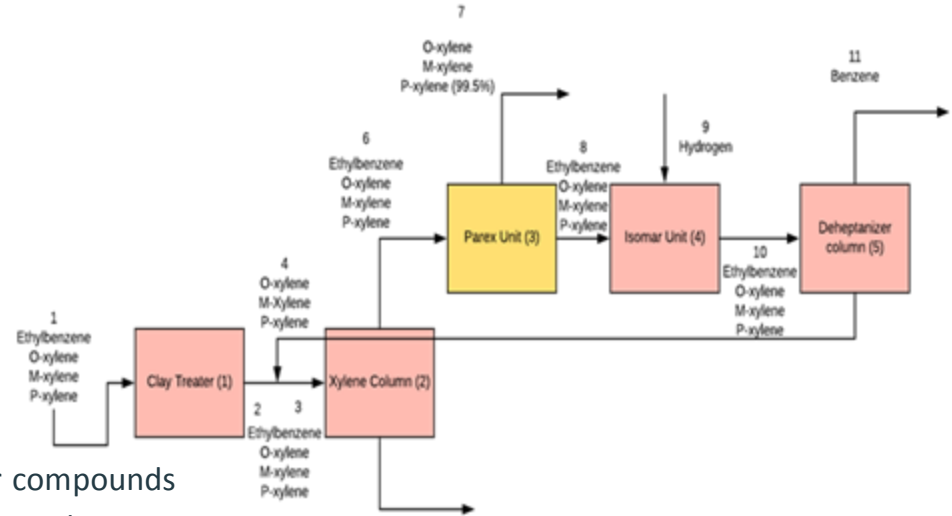
Process Description

Design Basis

- The basis of this plant is to produce 1.0 million metric tons of 99.5% pure *para*-xylene via the Parex separation in 350 operating days per calendar year.
- A majority of the *para*-xylene generated is demanded by the polyester industry(blankets, hats, bed sheets, water bottles, ropes, and safety belts) and more that half of polyester production takes place in China. Additionally, several *para*-xylene plants in China that can produce 1.0+ million metric tons of *para*-xylene are already operating.
- The new plant would be built in China with the intention of being competitive with other *para*-xylene producers as well as being accessible to the polyester industry that will purchase the product.

BFD:

- Aromatic blend feeds into clay treater to remove impurities.
- Next a xylene column separates heavier compounds from the feed and recycle stream leaving only C8 aromatics.
- The Parex unit Separates the final *para*-xylene product before sending the remaining stream to the isomar unit where *ortho*-xylene and *meta*-xylene are converted into *para*-xylene by isomerization.
- That stream then moves to a deheptanizer column which breaks down ethylbenzene and toluene into benzene which is then removed from the system.
- This leaves only xylene isomers which are recycled back to the stream leaving the clay treater and the xylenes are conserved within the system.



PFD:

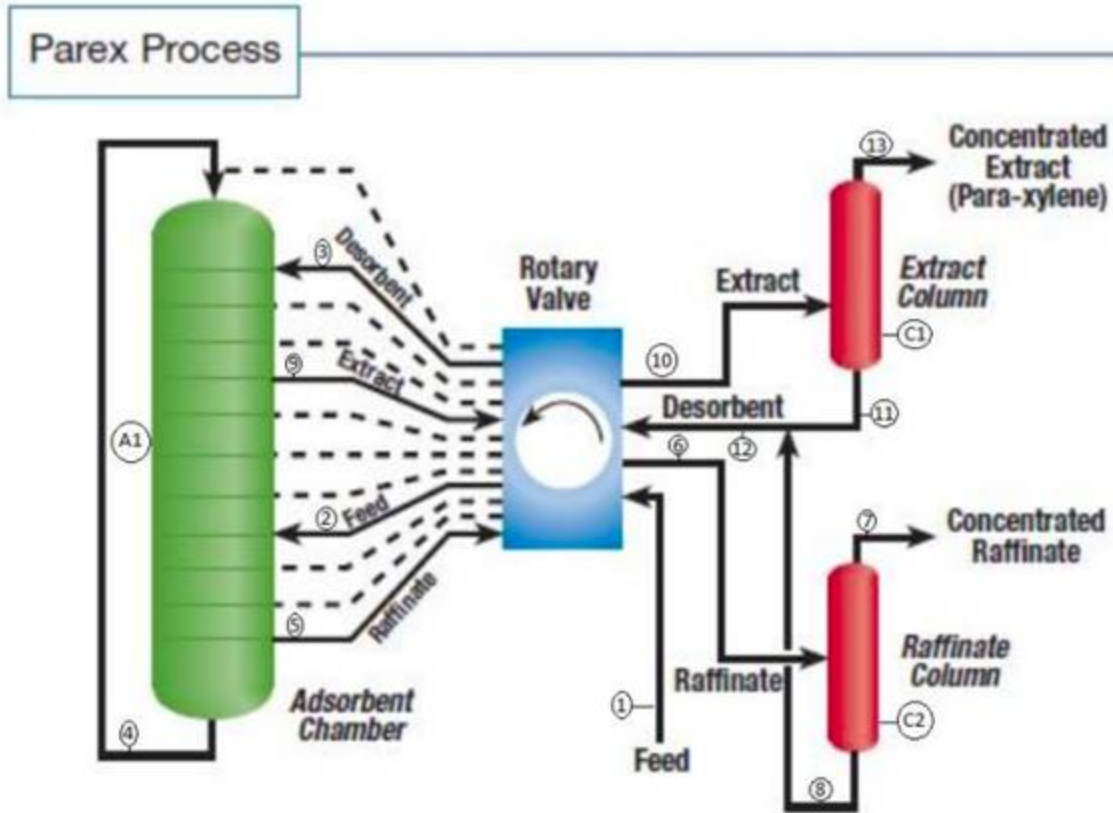


Figure 14. Process Flow Diagram (PFD- PAREX Unit) [16]

PAREX Unit Material Balance		Units	IN	OUT
EB	Ethylbenzene	kg/hr	88383.4	88383.4
P-x	Para-Xylene	kg/hr	148989.2	148989.2
O-x	Ortho-Xylene	kg/hr	80176.4	80176.4
M-x	Meta-xylene	kg/hr	313761	313761

Table 3. Parex input and output data.

$$m_6 = 631310 \frac{kg}{hr}$$

$$m_7 = 126262 \frac{kg}{hr}$$

$$m_8 = 505048 \frac{kg}{hr}$$

$$m_{6,EB} = 88383.4 \frac{kg}{hr}$$

$$m_{7,P-x} = 125630.7 \frac{kg}{hr}$$

$$m_{8,EB} = 88383.4 \frac{kg}{hr}$$

$$m_{6,P-x} = 148989.2 \frac{kg}{hr}$$

$$m_{7,M-x} = 505.0 \frac{kg}{hr}$$

$$m_{8,P-x} = 23232.2 \frac{kg}{hr}$$

$$m_{6,M-x} = 313761.1 \frac{kg}{hr}$$

$$m_{7,O-x} = 126.3 \frac{kg}{hr}$$

$$m_{8,M-x} = 313129.8 \frac{kg}{hr}$$

$$m_{6,O-x} = 80176.4 \frac{kg}{hr}$$

$$m_{8,O-x} = 80302.6 \frac{kg}{hr}$$



Thank you for listening!

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Additional Slides

